
Designing Forecasting Models for Rare High-Impact Events in Electricity Markets

Martin He
Carnegie Mellon University
Pittsburgh, PA
kunzeh@andrew.cmu.edu

Abstract

This project studies forecasting methods for electricity prices in the ERCOT market, with a focus on rare but economically significant price spikes. Classical baselines such as ARIMA are compared with deep learning approaches including vanilla LSTM, weighted loss LSTM, continuous weighting, and oversampling strategies. While ARIMA performs well in normal regimes, it fails to capture extreme events. Results show that naive LSTM models underpredict spikes, while aggressive spike-focused methods introduce excessive false positives. A fixed weighted LSTM achieves the best economic tradeoff by balancing spike detection and false alarm rates. Additionally, post-hoc threshold tuning does not meaningfully improve economic outcomes, highlighting that model training, rather than threshold selection, is the primary driver of performance. This work reframes spike forecasting as a cost-sensitive decision problem under heavy-tailed distributions, where model design must balance rare event detection against false alarm rates.

1 Motivation and Objectives

Electricity prices exhibit strong seasonality, weather dependence, and heavy-tailed spikes that make forecasting challenging[1]. Accurate prediction of extreme price movements is important not only for forecasting accuracy in the average case, but also for grid reliability, risk management, and energy trading decisions.

The central research question is:

How should forecasting models be designed to handle rare, high-impact events under asymmetric cost, where missing extreme events and overpredicting them have fundamentally different consequences?

2 Related Work

Electricity price forecasting is challenging due to non-stationarity, multiple seasonal patterns, weather dependence, and occasional extreme price spikes [4]. Classical statistical approaches such as ARIMA models remain strong baselines for short-horizon forecasting, particularly when seasonal and autoregressive structures dominate the dynamics [1].

Neural sequence models such as LSTM have been widely applied to time-series forecasting because they can capture nonlinear temporal dependencies. Prior studies have shown that LSTM-based approaches can outperform traditional machine learning methods, primarily in terms of aggregate error metrics such as MAE or RMSE [2], which may not fully capture economically critical behaviors such

as price spikes. More recent architectures such as N-BEATSx extend neural forecasting frameworks by incorporating exogenous covariates [3], which can improve performance in electricity price prediction where weather variables influence demand and price volatility

3 Methodology

3.1 Data Transformation and Preprocessing

Before fitting any models, the electricity price series is transformed using the inverse hyperbolic sine (asinh) transformation to reduce the impact of extreme price spikes while preserving sign and relative scale:

$$y_t = \text{asinh}(x_t) = \ln \left(x_t + \sqrt{x_t^2 + 1} \right).$$

Autocorrelation (ACF) and partial autocorrelation (PACF) plots are then examined to determine appropriate differencing. First-order differencing is applied to remove the overall trend, and an additional seasonal difference with lag 24 is used to account for daily seasonality in the hourly data.

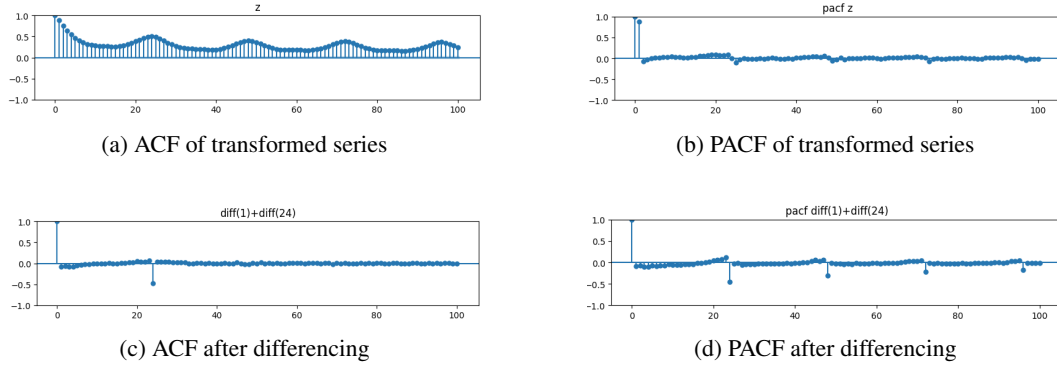


Figure 1: ACF and PACF diagnostics before and after differencing

The ACF of the raw series shows slow decay and strong seasonal peaks at multiples of 24 hours, indicating non-stationarity and daily periodicity. After applying first-order and seasonal differencing, the ACF decays rapidly with a single significant spike at lag 1 and a seasonal spike at lag 24, suggesting an MA(1) structure with seasonal MA(1) components. The PACF does not show a clear cutoff, indicating weak autoregressive effects.

The transformed and differenced series is used as the modeling target for both the ARIMA baseline and all deep learning models. This ensures a consistent comparison across methods, where each model operates on the same stationary representation of the underlying price dynamics. This design isolates differences in modeling capacity rather than differences in data preprocessing.

3.2 Baseline Model

$$\phi(B)(1 - B)^d y_t = \theta(B)\varepsilon_t$$

An ARIMA model is implemented as the primary statistical baseline. The model is trained on the transformed and differenced series using a walk-forward forecasting procedure, where each prediction is made based only on past observations[1]. This setup reflects realistic forecasting conditions and allows direct comparison with deep learning models under the same data representation.

3.3 Deep Learning Models

All deep learning models are trained on the same transformed and differenced time series used by the ARIMA baseline. This preprocessing ensures stationarity and allows the models to focus on learning temporal patterns and extreme deviations rather than raw price levels.

To address the imbalance between normal observations and rare spike events, several LSTM-based variants are implemented and systematically evaluated. Each model modifies the training objective or data distribution to improve sensitivity to extreme price movements.

- **Vanilla LSTM:** baseline sequence model trained using standard loss without any spike-aware adjustments.
- **Weighted LSTM:** a fixed weighting factor w is applied to errors corresponding to spike events during training. Multiple values were tested ($w \in \{3, 4, 5, 8, 10\}$) to explore the effect of increasing emphasis on extreme events. A weight of $w = 3$ is selected as the final configuration, as higher weights lead to instability and excessive false positives.
- **Continuous Weighting:** instead of using a fixed spike weight, the loss is scaled continuously based on the magnitude of the target variable. Scaling factors $\alpha \in \{0.5, 1, 2, 3, 5\}$ are evaluated. While larger values increase spike sensitivity, they also amplify noise and result in overprediction of spikes.
- **Oversampling:** spike events are oversampled during training to rebalance the dataset. Oversampling factors $\in \{2, 3, 4, 5, 6\}$ are tested. Although this approach improves recall of spike events, it introduces a large number of false positives at higher sampling rates.

These experiments are designed to explore the tradeoff between improving spike detection and maintaining stable predictions. The results show that moderate intervention (fixed weighting) provides a more effective balance than aggressive approaches such as high weighting or heavy oversampling. This study focuses not only on predictive accuracy, but on how different training strategies affect the tradeoff between rare event sensitivity and false positive control.

4 Dataset

4.1 Dataset

The study uses publicly available electricity price data from the ERCOT Houston Hub. The dataset consists of hourly locational marginal prices (LMPs), which capture the cost of supplying electricity at a specific location in the ERCOT wholesale market.

The dataset spans four years from January 2020 through December 2023. To preserve the temporal structure of the data, the series was divided chronologically into training, validation, and test sets:

- Training set: January 2020 – December 2021
- Validation set: January 2022 – December 2022
- Test set: January 2023 – December 2023

The series is transformed and differenced as described in Section 3.1 to ensure stationarity and consistency across all models.

The resulting dataset forms a stationary hourly time series suitable for ARIMA-based forecasting and subsequent comparison with deep learning models.

5 Loss Functions and Evaluation

5.1 Loss Functions

Deep learning models are trained using variants of regression loss functions designed to improve sensitivity to rare spike events.

The baseline model uses Mean Absolute Error (MAE):

$$\mathcal{L}_{\text{MAE}} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

To address class imbalance and emphasize extreme events, a weighted loss is introduced:

$$\mathcal{L}_{\text{weighted}} = \frac{1}{n} \sum_{t=1}^n w_t \cdot |y_t - \hat{y}_t|$$

where the weight w_t is defined as:

$$w_t = \begin{cases} w, & \text{if } |y_t| > q_{0.95} \\ 1, & \text{otherwise} \end{cases}$$

and $q_{0.95}$ is the 95th percentile threshold used to define spike events.

In addition, a continuous weighting scheme is explored, where the loss is scaled as a function of the magnitude of $|y_t|$, allowing larger deviations to receive higher penalties without a hard threshold.

Oversampling is also evaluated as an alternative approach, where spike observations are replicated during training to rebalance the dataset. Unlike weighted loss, this modifies the data distribution rather than the loss function.

5.2 Spike Definition

Let y_t denote the transformed and differenced price series. A spike event is defined using a data-driven threshold:

$$q_{0.95} = \text{Quantile}_{0.95}(|y_1|, |y_2|, \dots, |y_T|)$$

A spike occurs when:

$$|y_t| > q_{0.95}$$

This definition isolates rare extreme events while maintaining robustness to noise.

5.3 Evaluation Metrics

Model performance is evaluated using both regression and classification metrics.

Regression Metrics:

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}$$

To better capture tail behavior, evaluation is performed separately for two regimes:

- **Normal regime:** $|y_t| \leq q_{0.95}$
- **Spike regime:** $|y_t| > q_{0.95}$

Spike Detection Metrics:

Spike prediction is also evaluated as a binary classification task. Predicted spikes are defined by applying a threshold to model outputs.

- Precision
- Recall
- F1 score

These metrics quantify the tradeoff between false positives and false negatives.

5.4 Economic Cost

To align evaluation with real-world objectives, an economic cost metric is defined:

$$\text{Cost} = c_{\text{FN}} \cdot \text{FN} + c_{\text{FP}} \cdot \text{FP}$$

where:

- FN = false negatives (missed spikes)
- FP = false positives (false alarms)

with $c_{\text{FN}} > c_{\text{FP}}$, reflecting the higher penalty of missing a spike compared to issuing a false alarm.

This formulation reframes the problem as a cost-sensitive decision task rather than purely a prediction problem.

5.5 Threshold Scaling

To convert continuous predictions into spike classifications, a threshold scaling parameter k is introduced:

$$|\hat{y}_t| > k \cdot q_{0.95}$$

The parameter k controls the tradeoff between sensitivity and precision. Lower values increase recall but introduce more false positives, while higher values reduce false positives at the cost of missing true spikes.

The effect of threshold scaling is analyzed in the results section.

6 Results

This section presents the empirical evaluation of all models under the cost-sensitive framework defined above, with a focus on understanding tradeoffs between spike detection and false positive control.

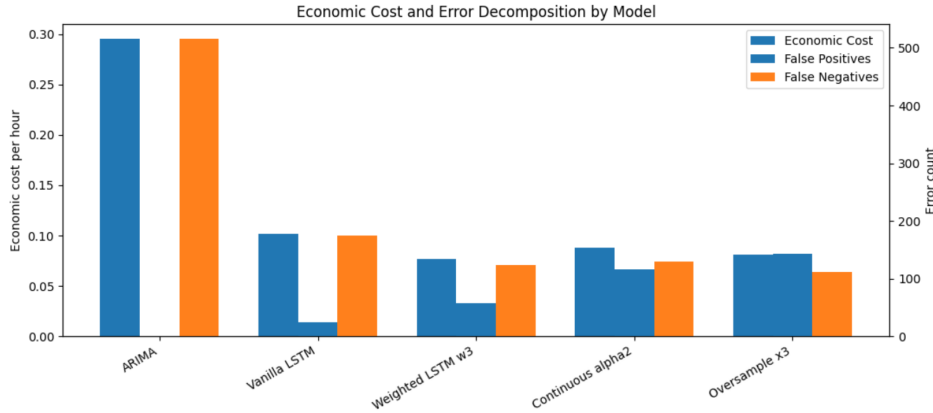


Figure 2: Economic cost and error decomposition by model. The weighted LSTM achieves the best tradeoff by reducing cost while maintaining a balanced false positive and false negative profile.

6.1 Overall Economic Performance

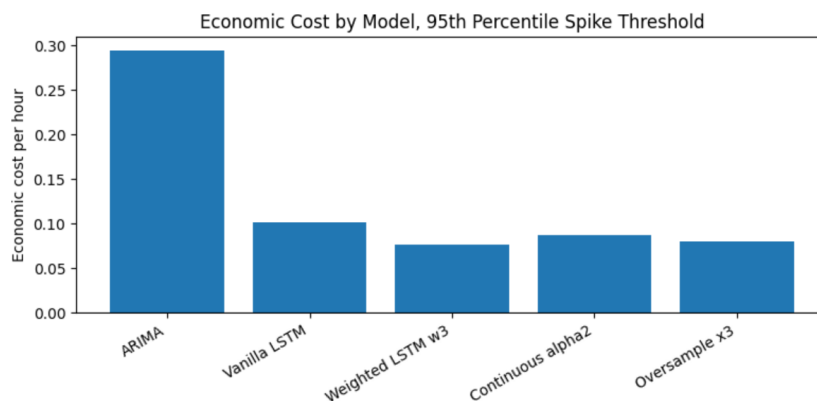


Figure 3: Economic cost by model (95th percentile threshold)

Economic cost provides the most meaningful evaluation metric, as spike events dominate financial outcomes. For each modeling approach, the best-performing configuration is selected based on validation performance and compared. The ARIMA baseline exhibits the highest cost due to its inability to capture extreme events. The vanilla LSTM significantly reduces cost but still underpredicts spikes. The weighted LSTM achieves the lowest economic cost, indicating the best balance between spike detection and false positives. Continuous weighting and oversampling increase spike sensitivity but introduce excessive false positives, leading to higher overall cost.

6.2 Spike Detection Performance

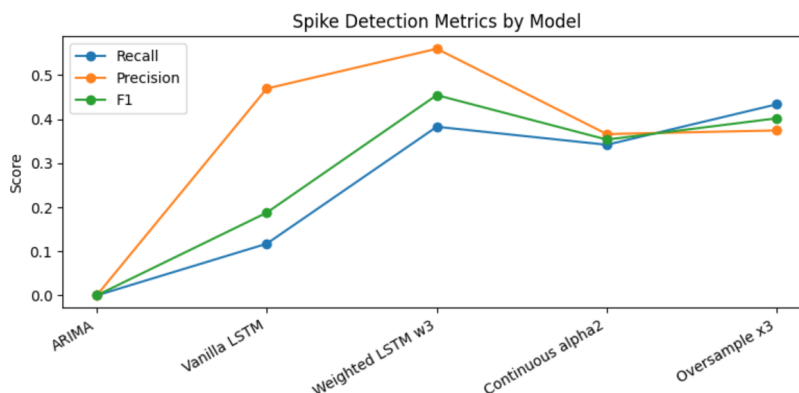


Figure 4: Precision, recall, and F1 score by model

ARIMA has near-zero spike detection capability. The vanilla LSTM improves precision but has low recall, indicating conservative predictions. The weighted LSTM achieves the best balance between precision and recall, resulting in the highest F1 score. Oversampling achieves high recall by increasing the representation of spike events during training, but suffers from reduced precision due to overfitting to rare patterns, resulting in excessive false positives.

6.3 False Positive and False Negative Tradeoff

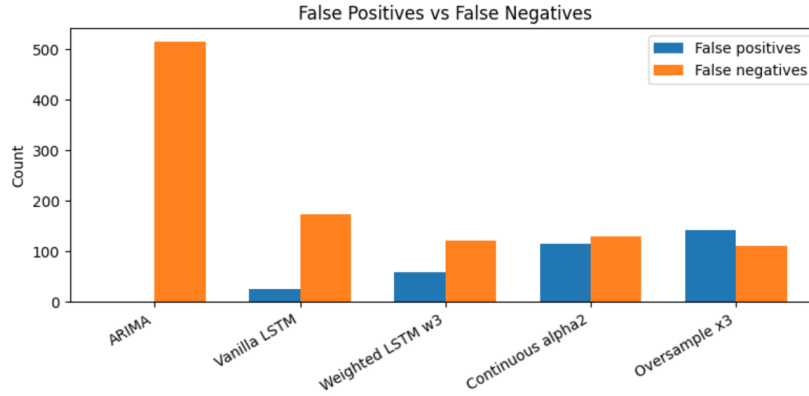


Figure 5: False positives and false negatives by model

ARIMA produces a large number of false negatives, missing most spike events. Oversampling reduces false negatives but introduces many false positives. The weighted LSTM achieves the most balanced tradeoff, explaining its superior economic performance.

6.4 Volatility Representation

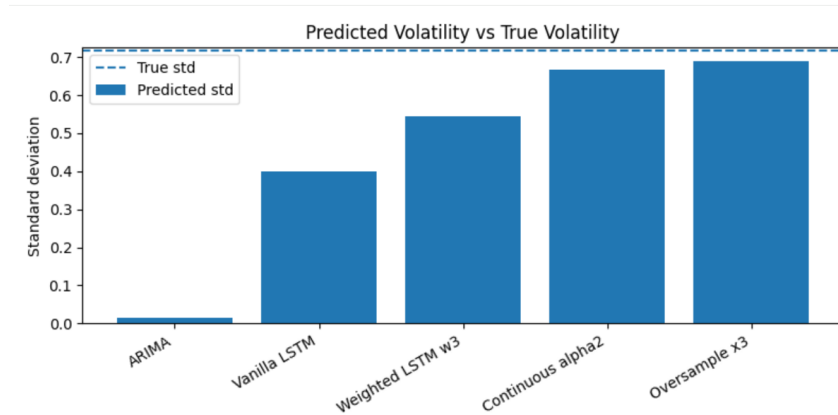


Figure 6: Predicted vs true volatility

ARIMA severely underestimates volatility, while the vanilla LSTM only partially improves this. Weighted and continuous models better approximate true volatility, while oversampling closely matches it but overcompensates. This demonstrates that capturing volatility alone is insufficient without controlling false positives.

6.5 Regime-Based Error

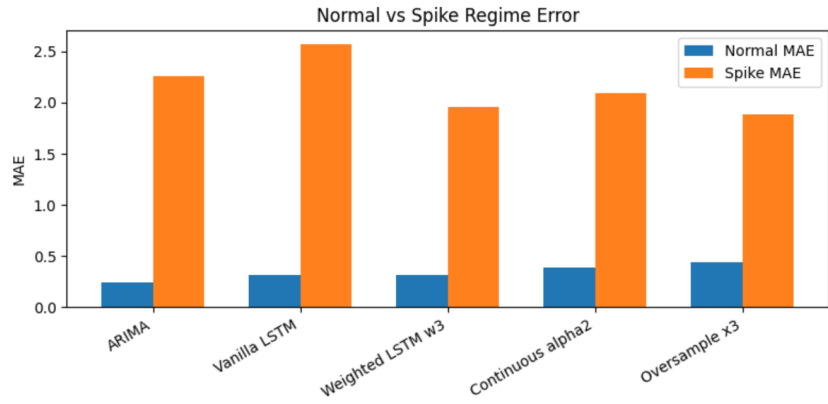


Figure 7: Normal vs spike MAE

All models perform similarly in the normal regime, while differences are substantial in the spike regime. The weighted LSTM reduces spike error without significantly harming normal performance. Oversampling achieves lower spike MAE but increases normal error and instability.

6.6 Threshold Scaling Analysis

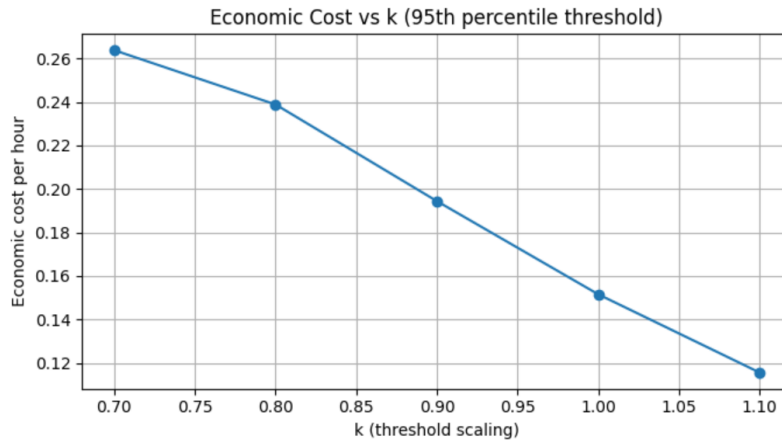


Figure 8: Economic cost vs threshold scaling k

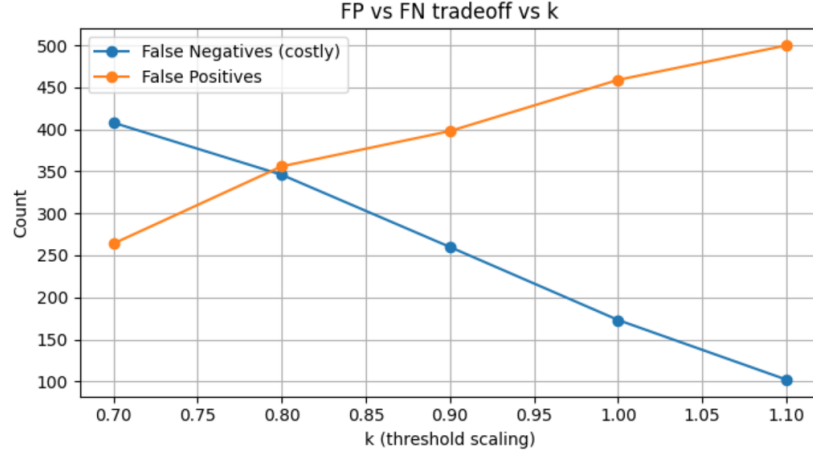


Figure 9: False positive vs false negative tradeoff vs k

Figure 8 and Figure 9 show how the threshold scaling parameter k affects economic cost and the tradeoff between false positives and false negatives. As k increases, the model becomes more conservative, reducing false positives and lowering economic cost. However, this comes at the expense of increased false negatives, meaning more true spike events are missed.

This highlights a key limitation of post-hoc threshold tuning: adjusting k shifts the balance between false positives and false negatives but does not improve the model’s underlying predictive capability.

These results indicate that performance is primarily determined during training rather than inference. The success of the weighted LSTM suggests that incorporating cost sensitivity directly into the training objective is more effective than relying on threshold adjustments after prediction.

6.7 Core Insight

A central finding of this study is that spike forecasting is governed by a tradeoff between sensitivity to rare events and stability under normal conditions. Models that aggressively prioritize spikes improve recall but introduce excessive false positives, while conservative models fail to capture extreme events. The weighted LSTM achieves the best balance, suggesting that moderate cost-sensitive training is more effective than either naive or extreme interventions.

7 Limitations and Robustness

Despite improved performance from deep learning models, electricity price forecasting remains fundamentally challenging due to the stochastic and externally driven nature of price dynamics. Extreme price spikes are often driven by exogenous factors such as weather shocks, demand surges, and market disruptions, many of which are only partially reflected in historical price data.

One limitation of this study is the omission of residual demand as an explicit feature. Residual demand, which represents the portion of total demand that must be met by marginal generators after accounting for renewable generation and other supply-side contributions, has been shown to more accurately explain electricity price formation [5]. By not incorporating this variable, the model may fail to capture key supply-side dynamics, particularly during periods of high renewable penetration or constrained generation capacity.

More broadly, system-level factors such as grid outages, transmission constraints, and generation availability can play a dominant role during extreme events and are not fully captured by price signals alone. Consequently, model performance during rare spikes may be limited by missing structural information that is difficult to infer purely from historical observations.

Another limitation arises from the inherent tradeoff between false positives and false negatives. Models that improve spike detection typically increase false alarm rates, while more conservative

models reduce false positives at the cost of missing critical events. Although the weighted LSTM achieves a favorable balance in this study, this tradeoff cannot be fully eliminated and reflects a fundamental constraint of decision-making under asymmetric misclassification costs [6].

In addition, the economic cost formulation adopted here simplifies real-world decision-making by assigning fixed penalties to false positives and false negatives. In practice, these costs are context-dependent and vary with market conditions, spike magnitude, risk exposure, and operational constraints. Future work could incorporate dynamic or context-aware cost functions to better reflect these complexities.

Finally, the evaluation is conducted on a single market (ERCOT Houston Hub), and the results may not directly generalize to other electricity markets with different regulatory structures, generation mixes, or demand profiles.

Despite these limitations, the consistency of results across multiple model variants suggests that the observed tradeoffs between spike sensitivity and false positive control are robust. These findings reflect fundamental properties of forecasting under heavy-tailed distributions, where rare extreme events dominate economic outcomes and require explicit modeling of both system conditions and decision objectives.

8 Discussion and Contribution

This study shows that electricity price forecasting for rare, high-impact events is fundamentally a decision-making problem under asymmetric cost, rather than purely a prediction task. Standard metrics such as RMSE or MAE fail to capture performance in this setting, as extreme price spikes dominate economic outcomes. By introducing an explicit cost formulation, this work reframes forecasting as optimizing decision quality, balancing the tradeoff between false negatives and false positives.

Empirical results reveal a consistent tradeoff across models. ARIMA fails to capture extreme events, resulting in high false negatives. Vanilla LSTM improves average performance but remains conservative. Aggressive methods such as oversampling and continuous weighting increase spike sensitivity but introduce excessive false positives. A fixed weighted LSTM achieves the best balance, suggesting that moderate cost-sensitive training is more effective than extreme interventions.

A key finding is that performance is primarily determined during training rather than inference. Threshold tuning can shift the balance between false positives and false negatives, but does not improve underlying predictive capability. This highlights the importance of incorporating cost sensitivity directly into the training objective.

Finally, the results suggest that price spikes are driven by underlying system conditions such as supply-demand imbalance and capacity constraints, rather than purely stochastic behavior. While this study focuses on historical price data, incorporating structural variables such as residual demand, outages, and generation capacity remains an important direction for future work.

Overall, this work contributes a decision-aware framework for electricity price forecasting under heavy-tailed distributions, demonstrating how model design influences the tradeoff between rare event detection and false positive control.

9 Bibliography

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